



MDP 251s

Metal Welding

Reports &

Assignments

Code & Sec. No.	
Student Name:	



[1] Basic Concepts

Define the Welding Process

Discuss briefly (in definite points) WPS, PQR, & WQTR



[2] Welding Joints

Sketch the different Welding Positions			
Flat	Horizontal	Vertical	Overhead

Sketch the different Types of Welding Joints				
Butt Joint	Lab Joint	Tee Joint	Corner Joint	Edge Joint

Sketch Fillet Weld & Illustrate the weld details	
1- Weld Throat 2- Weld Face 3- Weld Root 4- Weld Toe 5- Depth of Penetration 6- Weld Leg [SIZE]	

Sketch Groove Weld & Illustrate the weld details	
1- Groove Angle 2- Bevel Angle 3- Root Face 4- Root Opining 5- Groove Face	



Illustrate The Weld Joint Stresses [Normal & Shear Stresses].
[With aid of neat sketches]

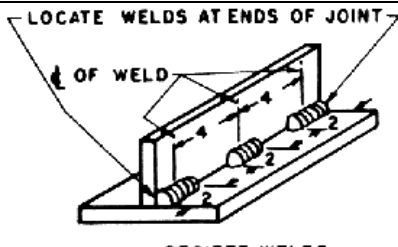
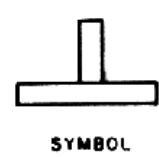
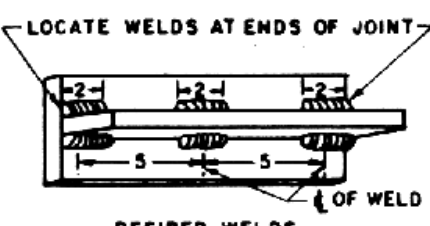
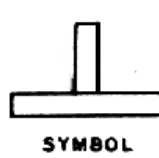
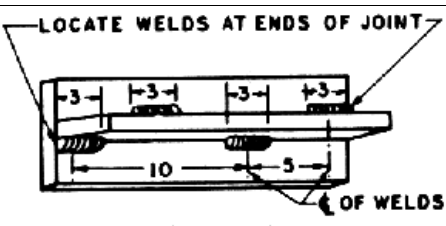
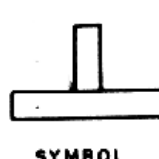
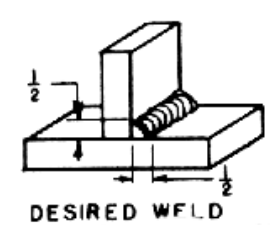
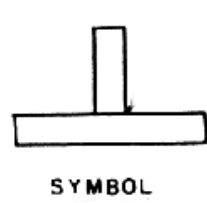
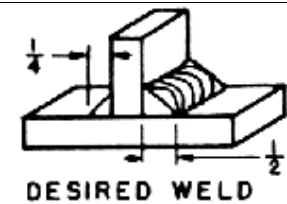
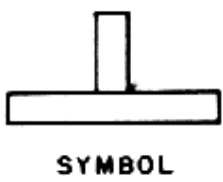
Normal Stresses

Shear Stresses

[3] Welding Symbols

Sketch & Explain the Elements of Welding Symbol

Assign the welding symbols:

1	 
2	 
3	 
4	 
5	 



[4] Welding Processes

Illustrate (with aid of neat sketch) the principles of **OAW** Welding Process.

Illustrate (with aid of neat sketch) the principles of **TW** Welding Process.



Illustrate (with aid of neat sketch) the principles of **SMAW** Welding Process.

Illustrate (with aid of neat sketch) the principles of **GMAW** Welding Process.



Illustrate (with aid of neat sketch) the principles of **GTAW** Welding Process.

Illustrate (with aid of neat sketch) the principles of **PAW** Welding Process.



Illustrate (with aid of neat sketch) the principles of **SW** Welding Process.

Illustrate (with aid of neat sketch) the principles of **SRW** Welding Process.



Illustrate (with aid of neat sketch) the principles of **EXW** Welding Process.

Illustrate (with aid of neat sketch) the principles of **FRW** Welding Process.



Compare between RSW & SFW processes.

[Arrange your answer in a table, with aid of neat sketch].

Compare between GMAW & PAW processes.

[Arrange your answer in a table]

[5] Arc Welding Heat Flow

Apply Rosenthal's Three-Dimensional Equation **YOU CAN SOLVE IT USING EXCEL SHEET & INSERT IT!**

- 1- **Calculate** the temperature of the proposed 10 points shown in the table.
- 2- **Plot** the Thermal cycle for the [2mm] line apart from the weld centerline.

Materials Data:

Stainless Steel [SS 304]

Thermal Conductivity

$K = 24.9$ [W/ m K]

Thermal Diffusivity

$\alpha = 5.3 \times 10^{-6}$ [m²/s]

Welding Process Data:

SMAW Process

$\eta = 80$ [%]

Welding Speed

$V = 150$ [mm/min]

Welding Voltage

$E = 20$ [Volt]

Welding Current

$I = 120$ [Amp.]

Preheating Condition:

100 [°C] Preheat

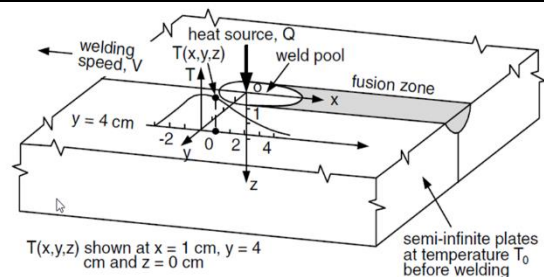
$T_0 = 373.15$ [K]

Welding Speed

$V = [\quad]$ [m/s]

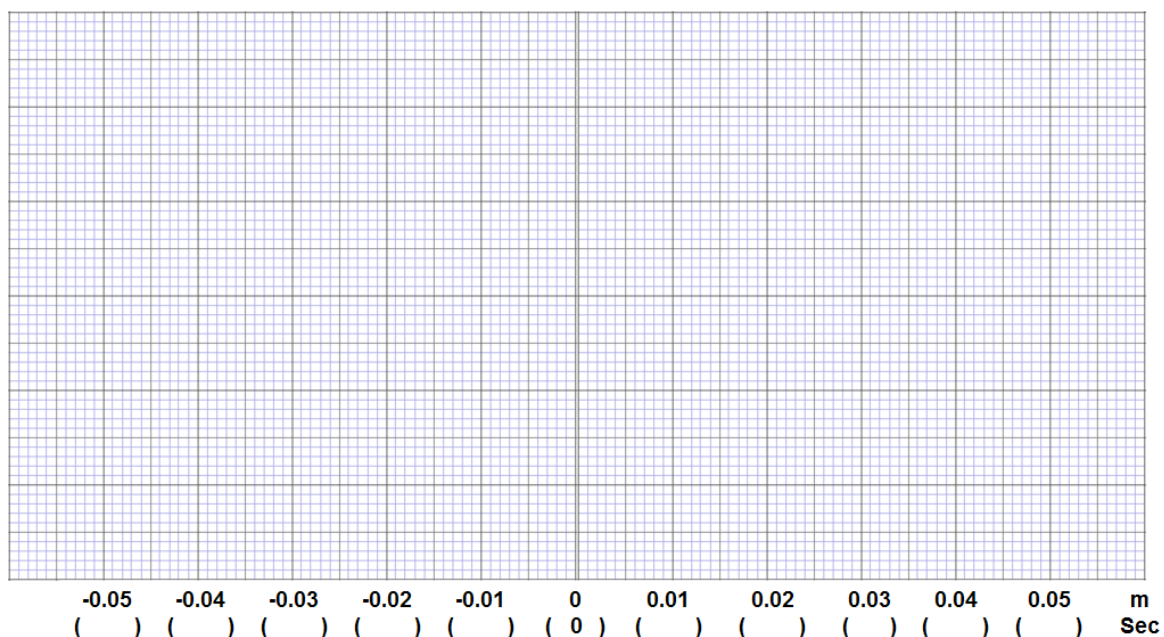
Rate of Heat Source

$Q = [\quad]$ [Watt]



$$\frac{2\pi(T - T_0)kR}{Q} = \exp\left[\frac{-V(R - x)}{2\alpha}\right]$$

	X	Y	Z	R	R-X	exp. $[-v(R-x)/2\alpha]$	$Q/(2\pi KR)$	T	T	Time
Points	[m]	[m]	[m]	[m]	[m]	Dimensionless	[K]	[K]	[°C]	[Sec]
1	-0.050	0.002	0							
2	-0.030	0.002	0							
3	-0.010	0.002	0							
4	-0.005	0.002	0							
5	-0.002	0.002	0							
6	0.002	0.002	0							
7	0.005	0.002	0							
8	0.010	0.002	0							
9	0.030	0.002	0							
10	0.050	0.002	0							





[6] Arc Welding Fluid Flow

Compare between **Sharp Electrode & Flat-End Electrode**. [With aid of neat sketch].
 The effect on both **Lorentz force and Fluid flow** in Welding Arcs

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Explain the **Driving Forces for Fluid Flow** in Weld Pools

1-	Sketch
2-	Sketch
3-	Sketch
4-	Sketch

[7] Metal Welding Solidification

Explain Variations in **G** & **R** along pool boundary

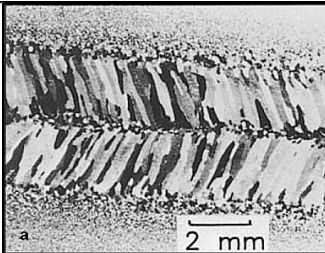
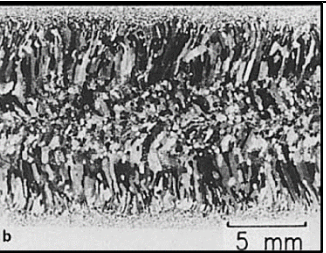
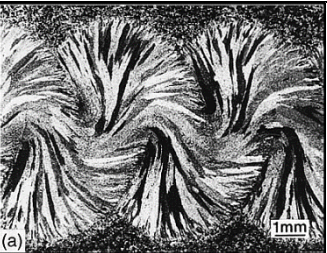
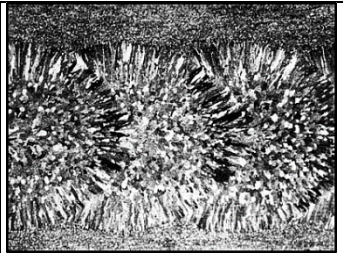
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Explain the Effect of **G** & **R** on Solidification Structure & Size

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Explain your opinion in the shown weld Macrographs

(Use Definite Words for the used method for Grain Structure Control)



Explain The Solidification Cracking Mechanism

with aid of neat sketch

Explain The Liquation Cracking Mechanism

with aid of neat sketch



[8] Welding Defects

Defect	Schematic Sketch	Major Causes	Action for Prevention
Porosities			
Slag Inclusions			
Under-cut			
Under-Fill			
LOF			
LOP			
Hot Cracks			
Lamellar Cracks			



[9] Welding Inspection

Explain briefly the **Liquid penetrant** method, and mention its applications.

Sketch

Explain briefly **MPT nondestructive method**, and mention its applications.

Sketch

Illustrate (**with aid of neat sketch**) Ultrasonic inspection **UT** method

Sketch



[10] Calculations of Cracking Susceptibility

On one hand, the susceptibility to Hot cracking, is based on weld composition, and can be estimated with empirical equations, such as: Unit of Cracking Susceptibility

$$UCS = 230 C + 190 S + 75 P + 45 Nb - 12.3 Si - 5.4 Mn - 1,$$

On the other hand, Graville diagram shows susceptibility of steels to hydrogen-assisted cold cracking relative to carbon content and carbon equivalent.

$$CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15$$

1. Redraw and Illustrate Graville diagram.
2. Calculate the **UCS** for the following steel alloys and define their susceptibility to Hot cracking.
3. Calculate the **CE** for the following steel alloys and define their susceptibility to Cold cracking.

Steel / Composition	C%	Si %	Mn%	Cr %	Ni %	Mo%	V%	P %	S %
DIN C22 (Structural Steel)	0.22	0.40	0.45	----	----	----	----	0.045	0.045
DIN 38Si6 (Spring Steel)	0.38	1.50	0.60	0.20	0.20	0.10	----	0.050	0.050
DIN 35NiCr18 (Heat-treatable Steel)	0.30	0.30	0.60	1.50	4.50	----	----	0.035	0.035
DIN C 85 W (Carbon Tool Steel)	0.85	0.35	0.65	1.20	----	----	----	0.025	0.020

Arrange your answer in the table. (Use **definite words**, such as **High, Medium, and Low risk**)

Steel	CE	Graville diagram Zone	Susceptibility to Cold Cracking	UCS	Susceptibility to Hot Cracking

Draw Graville diagram & Plot the different Steels